

Optimizing the Deployment of Tiny Transformers on Low-Power MCUs

Victor Jean-Baptiste Jung , *Graduate Student Member, IEEE*, Alessio Burrello ,
Moritz Scherer , *Member, IEEE*, Francesco Conti , *Member, IEEE*, and Luca Benini , *Fellow, IEEE*

Abstract—Transformer networks are rapidly becoming State of the Art (SotA) in many fields, such as Natural Language Processing (NLP) and Computer Vision (CV). Similarly to Convolutional Neural Networks (CNNs), there is a strong push for deploying Transformer models at the extreme edge, ultimately fitting the tiny power budget and memory footprint of Micro-Controller Units (MCUs). However, the early approaches in this direction are mostly ad-hoc, platform, and model-specific. This work aims to enable and optimize the flexible, multi-platform deployment of encoder Tiny Transformers on commercial MCUs. We propose a complete framework to perform end-to-end deployment of Transformer models onto single and multi-core MCUs. Our framework provides an optimized library of kernels to maximize data reuse and avoid unnecessary data marshaling operations into the crucial attention block. A novel Multi-Head Self-Attention (MHSA) inference schedule, named Fused-Weight Self-Attention (FWSA), is introduced, fusing the linear projection weights offline to further reduce the number of operations and parameters. Furthermore, to mitigate the memory peak reached by the computation of the attention map, we present a Depth-First Tiling (DFT) scheme for MHSA tailored for cache-less MCU devices that allows splitting the computation of the attention map into successive steps, never materializing the whole matrix in memory. We evaluate our framework on three different MCU classes exploiting ARM and RISC-V Instruction Set Architecture (ISA), namely the STM32H7 (ARM Cortex M7), the STM32L4 (ARM Cortex M4), and GAP9 (RV32IMC-XpulpV2). We reach an average of $4.79 \times$ and $2.0 \times$ lower latency compared to SotA libraries CMSIS-NN (ARM) and PULP-NN (RISC-V), respectively. Moreover, we show that our MHSA depth-first tiling

scheme reduces the memory peak by up to $6.19 \times$, while the fused-weight attention can reduce the runtime by $1.53 \times$, and number of parameters by 25%. Leveraging the optimizations proposed in this work, we run end-to-end inference of three SotA Tiny Transformers for three applications characterized by different input dimensions and network hyperparameters. We report significant improvements across the networks: for instance, when executing a transformer block for the task of radar-based hand-gesture recognition on GAP9, we achieve a latency of 0.14 ms and energy consumption of 4.92 μ J, $2.32 \times$ lower than the SotA PULP-NN library on the same platform.

Index Terms—Deep neural networks, transformers, micro-controller units, edge computing, DNN acceleration.

I. INTRODUCTION

IN the last few years, there has been a significant trend towards moving computing workload from centralized facilities towards the extreme edge of the Internet of Things (IoT), improving privacy, efficiency, and ensuring a predictable and constant latency, independent from the network coverage and congestion [1]. As a consequence, modern IoT endpoints have been changing from simple Micro-Controller Units (MCUs) equipped with the sensor infrastructure for data collection and transmission towards more complex heterogeneous Systems-on-Chip (SoCs) characterized by dedicated accelerators and enhanced Instruction Set Architectures (ISAs) to optimize efficient on-board computations – while still meeting tight power, performance, and cost constraints. For this reason, extensive research has been conducted on how to squeeze complex Machine Learning (ML) models into such constrained devices with a strong emphasis on Deep Neural Networks (DNNs) due to their success in solving many challenging tasks. Standard hardware-agnostic optimization techniques include topological changes [2], data quantization [3], [4], and pruning [5]. These efforts are often collectively labeled as *TinyML* and are applied to State of the Art (SotA) Convolutional Neural Networks (CNNs), such as ResNets [6] or MobileNets [7], that dominates most of the tasks of the IoT domain, including image recognition, object detection, or time-series analysis.

At the same time, increasing efforts have been dedicated to finding a successor architecture to CNNs for IoT processing to increase tasks' accuracy further. A strong candidate is the Transformer architecture [8], which first emerged as SotA model in Natural Language Processing (NLP) and is now also SotA in other domains such as Computer Vision (CV) and

Received 28 March 2024; revised 19 September 2024; accepted 25 October 2024. Date of publication 18 November 2024; date of current version 20 January 2025. This work was supported in part by the Chips Joint Undertaking (Chips-JU) TRISTAN Project under Grant Agreement 101095947, and in part by the ConVolve Project under Grant Agreement 101070374. The JU and CONVOLVE receive support from the European Union's Horizon Europe Research and Innovation Program. Recommended for acceptance by R. Bunescu. (Corresponding author: Victor Jean-Baptiste Jung.)

Victor Jean-Baptiste Jung and Moritz Scherer are with the Integrated Systems Laboratory (IIS), ETH Zürich, ETZ, 8092 Zürich, Switzerland (e-mail: jungvi@iis.ee.ethz.ch; scheremo@iis.ee.ethz.ch).

Alessio Burrello is with the Department of Electrical, Electronic and Information Engineering, University of Bologna, 40136 Bologna, Italy, and also with the Interuniversity Department of Regional and Urban Studies and Planning, Politecnico di Torino, 10129 Turin, Italy (e-mail: alessio.burrello@polito.it).

Francesco Conti is with the Department of Electrical, Electronic and Information Engineering, University of Bologna, 40136 Bologna, Italy (e-mail: f.conti@unibo.it).

Luca Benini is with the Integrated Systems Laboratory (IIS), ETH Zürich, ETZ, 8092 Zürich, Switzerland, and also with the Department of Electrical, Electronic and Information Engineering, University of Bologna, 40136 Bologna, Italy (e-mail: lbenini@iis.ee.ethz.ch; luca.benini@unibo.it).

Digital Object Identifier 10.1109/TC.2024.3500360

Original p. 2 is not reproduced in this excerpt.
Full article: IEEE Xplore, <https://doi.org/10.1109/TC.2024.3500360> (© IEEE).

Original p. 3 is not reproduced in this excerpt.
Full article: IEEE Xplore, <https://doi.org/10.1109/TC.2024.3500360> (© IEEE).

Original p. 4 is not reproduced in this excerpt.
Full article: IEEE Xplore, <https://doi.org/10.1109/TC.2024.3500360> (© IEEE).

Original p. 5 is not reproduced in this excerpt.
Full article: IEEE Xplore, <https://doi.org/10.1109/TC.2024.3500360> (© IEEE).

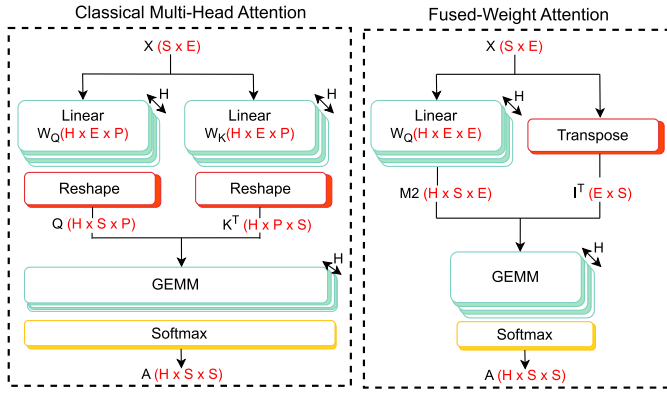


Fig. 3. Diagram of the Classical Attention used in the MHSA and the proposed Fused-Weight Attention. The dimension of each tensor is specified in red.

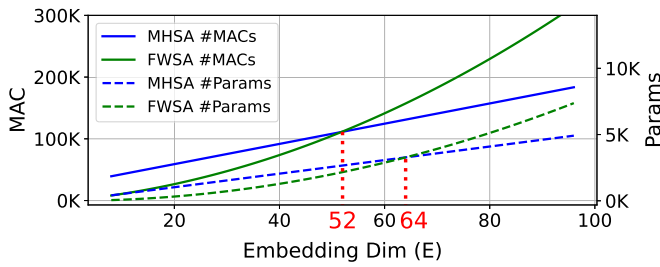


Fig. 4. Number of parameters and Multiply-Accumulate (MAC) as a function of the embedding dimension E and for $S = 32$, $P = 32$ and $H = 8$. The intersection points happen at $E = 52$ and $E = 64$ for the number of MAC and the number of parameters, respectively.

$$A = \phi(X \cdot W_Q \cdot (X \cdot W_K)^T) \quad (4)$$

$$A = \phi(X \cdot W_Q \cdot W_K^T \cdot X^T) \quad (5)$$

$$A = \phi(X \cdot W^* \cdot X^T) \quad (6)$$

In the equations above, we reorder the operations using the associativity of the matrix product and the transposition's reversal order of the product. The fused weight matrix W^* is computed offline and considered as fixed parameters during the inference. In detail, we reduce the number of matrix multiplications to perform online from three to two. Fig. 3 shows the computational graph of the classical Multi-Head Attention (left) and the FWSA (right), we indicate the dimension of the tensors in red.

To evaluate when this transformation is beneficial, we compare the number of operations O and parameters θ between the FWSA and the MHSA in Eq. 7 & 8 below:

$$\begin{cases} O_{MHSA} = 2HSPE + HS^2P \\ O_{FWSA} = HSE^2 + HS^2E \end{cases} \quad (7)$$

$$\begin{cases} \theta_{MHSA} = 2HPE \\ \theta_{FWSA} = HE^2 \end{cases} \quad (8)$$

Looking at Eq. 7 & 8, we notice that using the FWSA is indeed beneficial for some conditions of S , P , and E values. More specifically, in Eq. 9, we provide the inequalities that need

to be satisfied to benefit from the FWSA in terms of number of operations (left) and parameters (right).

$$E < P - \frac{S}{2} + \frac{\sqrt{4P^2 + S^2}}{2} \quad E < 2P \quad (9)$$

In Fig. 4, we show the relationship between E and the number of MAC/Parameters for the MHSA and FWSA. For $E < 52$, FWSA always shows a lower number of MAC and parameters. When considering the intermediate condition of $52 \leq E < 64$, the FWSA reduces the number of parameters but increases the number of MACs. Finally, in the third case ($E \geq 64$), the MHSA is more advantageous than the FWSA.

C. Transformers Quantization With QuantLib

To quantize Transformers, we employ and extend the open-source quantization library QuantLib. QuantLib transforms the ONNX graph representing the floating-point model of a DNN to an integerized ONNX graph. This transformation from a floating-point to an integerized graph is done in two steps.

All operation nodes in the graph, such as matrix multiplication, convolution, or softmax, are replaced by their integer equivalent. As the softmax, GELU, and layer normalization present in Transformers do not have an integer equivalent, we extend QuantLib to support the integer version of these operands from I-BERT.

Then, we add a re-quantization step after operations whose output is not represented in 8 bit. For instance, the output type of an integer GEMM is `int32`; thus, to convert it to 8 bit, we have to re-quantize it. We use the uniform symmetric quantizer from the SotA Trained Quantization Threshold (TQT) method [3]. Below, we describe the functional behavior of the TQT quantizer denoted ψ :

$$\psi(x) = \max(2^{b-1}, \min((x \cdot \epsilon_{mul}) \gg \epsilon_{div}, 2^{b-1} - 1)) \quad (10)$$

The number of bits of the quantized values is b , while ϵ_{mul} and ϵ_{div} are the re-quantization parameters. This quantizer constrains the scale factor to a powers-of-2 (i.e., $2^{\epsilon_{div}}$) and uses per-tensor scaling of activations and weights, making it hardware-friendly as we can use the bit-shift operation to apply ϵ_{div} . Additionally, we use Quantization Aware Training (QAT) to adjust weights to reduce the accuracy loss induced by quantization. Note that we could use Post-Training Quantization (PTQ) instead of QAT, we choose to use QAT because it mitigates even further the quantization accuracy loss, compared to PTQ.

After applying the necessary transformations and obtaining an integerized ONNX graph, we integrate into QuantLib a new export module to format the graph such that it can be ingested by the deployment framework (DORY or DumpO).

D. Deployment Framework Integration

For ARM platforms, we use DumpO to compile the integerized ONNX graph into C code. DumpO first transforms the graph so that each node can be executed by one kernel from

Original p. 7 is not reproduced in this excerpt.
Full article: IEEE Xplore, <https://doi.org/10.1109/TC.2024.3500360> (© IEEE).

Original p. 8 is not reproduced in this excerpt.
Full article: IEEE Xplore, <https://doi.org/10.1109/TC.2024.3500360> (© IEEE).

Original p. 9 is not reproduced in this excerpt.
Full article: IEEE Xplore, <https://doi.org/10.1109/TC.2024.3500360> (© IEEE).

Original p. 10 is not reproduced in this excerpt.
Full article: IEEE Xplore, <https://doi.org/10.1109/TC.2024.3500360> (© IEEE).

Original p. 11 is not reproduced in this excerpt.
Full article: IEEE Xplore, <https://doi.org/10.1109/TC.2024.3500360> (© IEEE).

Original p. 12 is not reproduced in this excerpt.
Full article: IEEE Xplore, <https://doi.org/10.1109/TC.2024.3500360> (© IEEE).

Original p. 13 is not reproduced in this excerpt.
Full article: IEEE Xplore, <https://doi.org/10.1109/TC.2024.3500360> (© IEEE).

Original p. 14 is not reproduced in this excerpt.
Full article: IEEE Xplore, <https://doi.org/10.1109/TC.2024.3500360> (© IEEE).

VI. CONCLUSION AND FUTURE WORK

In this work, we proposed an end-to-end flow to enable efficient deployment of small Transformer models onto commercial MCUs. Our kernel library, tailored for MHSA, together with our optimized schedule and tiling strategy, allows us to speed up the execution of the attention block by a factor of $2.94\times$ on average on RISC-V and ARM platforms. Furthermore, we demonstrate the efficiency of our flow by deploying three Tiny Transformers onto the GAP9 MCU, reaching an average energy consumption and latency of $202\mu\text{J}$ and 5.92 ms , respectively. Our work is open-source at <https://github.com/pulp-platform/pulp-transformer>. Future research direction would extend our framework to support emerging transformers such as encoder-decoder and decoder-only.

ACKNOWLEDGMENT

The authors thank Marco Fariselli from GreenWaves Technologies for the insightful discussions, particularly on the FWSA strategy.

REFERENCES

- [1] C. R. Banbury et al., "Benchmarking TinyML systems: Challenges and direction," 2020, *arXiv:2003.04821*.
- [2] J. Lin et al., "MCUNet: Tiny deep learning on IoT devices," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 33, 2020, pp. 11711–11722.
- [3] S. Jain, A. Gural, M. Wu, and C. Dick, "Trained quantization thresholds for accurate and efficient fixed-point inference of deep neural networks," *Proc. Mach. Learn. Syst.*, vol. 2, 2020, pp. 112–128.
- [4] C. Gong, Y. Chen, Y. Lu, T. Li, C. Hao, and D. Chen, "VecQ: Minimal loss DNN model compression with vectorized weight quantization," *IEEE Trans. Comput.*, vol. 70, no. 5, pp. 696–710, May 2021.
- [5] S. Han, H. Mao, and W. J. Dally, "Deep compression: Compressing deep neural network with pruning, trained quantization and Huffman coding," in *Proc. 4th Int. Conf. Learn. Representations (ICLR)*, San Juan, Puerto Rico, 2016.
- [6] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2016, pp. 770–778.
- [7] A. G. Howard et al., "MobileNets: Efficient convolutional neural networks for mobile vision applications," 2017, *arXiv:1704.04861*.
- [8] A. Vaswani et al., "Attention is all you need," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 30, 2017, pp. 6000–6010.
- [9] A. Burrello, M. Scherer, M. Zangheri, F. Conti, and L. Benini, "A microcontroller is all you need: Enabling transformer execution on low-power IoT endnodes," in *Proc. IEEE Int. Conf. Omni-Layer Intell. Syst. (COINS)*, 2021, pp. 1–6.
- [10] A. Burrello, A. Garofalo, N. Bruschi, G. Tagliavini, D. Rossi, and F. Conti, "DORY: Automatic end-to-end deployment of real-world DNNs on low-cost IoT MCUs," *IEEE Trans. Comput.*, vol. 70, no. 8, pp. 1253–1268, Aug. 2021.
- [11] D. Hendrycks and K. Gimpel, "Gaussian error linear units (GELUs)," 2016, *arXiv:1606.08415*.
- [12] A. Garofalo, M. Rusci, F. Conti, D. Rossi, and L. Benini, "PULP-NN: Accelerating quantized neural networks on parallel ultra-low-power RISC-V processors," *Philos. Trans. Roy. Soc. A*, vol. 378, no. 2164, 2020, Art. no. 20190155.
- [13] A. Katharopoulos, A. Vyas, N. Pappas, and F. Fleuret, "Transformers are RNNs: Fast autoregressive transformers with linear attention," in *Proc. Int. Conf. Mach. Learn.*, PMLR, 2020, pp. 5156–5165.
- [14] K. M. Choromanski et al., "Rethinking attention with performers," in *Proc. 9th Int. Conf. Learn. Representations (ICLR)*, Virtual Event, Austria, OpenReview.net, 2021, pp. 63–101.
- [15] D. Bolya, C.-Y. Fu, X. Dai, P. Zhang, and J. Hoffman, "Hydra attention: Efficient attention with many heads," in *Proc. Eur. Conf. Comput. Vis.*, Switzerland: Springer Nature, 2022, pp. 35–49.
- [16] B. Peng et al., "RWKV: Reinventing RNNs for the transformer era," in *Findings Assoc. Comput. Linguistics: EMNLP*, H. Bouamor, J. Pino, and K. Bali, Eds., Singapore: Association for Computational Linguistics, Dec. 2023, pp. 14048–14077.
- [17] S. Wang, B. Z. Li, M. Khabsa, H. Fang, and H. Ma, "Linformer: Self-attention with linear complexity," 2020, *arXiv:2006.04768*.
- [18] Z. Qin et al., "The devil in linear transformer," in *Proc. Conf. Empirical Methods Natural Lang. Process.*, Abu Dhabi, United Arab Emirates: Association for Computational Linguistics, 2022, pp. 7025–7041.
- [19] N. Shazeer, "Fast transformer decoding: One write-head is all you need," 2019, *arXiv:1911.02150*.
- [20] J. Ainslie, J. Lee-Thorp, M. de Jong, Y. Zemlyanskiy, F. Lebron, and S. Sanghai, "GQA: Training generalized multi-query transformer models from multi-head checkpoints," in *Proc. Conf. Empirical Methods Natural Lang. Process.*, H. Bouamor, J. Pino, and K. Bali, Eds., Singapore: Association for Computational Linguistics, Dec. 2023, pp. 4895–4901.
- [21] H. Touvron et al., "Llama 2: Open foundation and fine-tuned chat models," 2023, *arXiv:2307.09288*.
- [22] T. Dao, "FlashAttention-2: Faster attention with better parallelism and work partitioning," in *Proc. 12th Int. Conf. Learn. Representations*, 2024, pp. 210–224.
- [23] W. Kwon et al., "Efficient memory management for large language model serving with PagedAttention," in *Proc. 29th Symp. Operating Syst. Princ. (SOSP '23)*, New York, NY, USA: ACM, 2023, pp. 611–626.
- [24] R. Pope, "Efficiently scaling transformer inference," *Proc. Mach. Learn. Syst.*, vol. 5, 2023, pp. 606–624.
- [25] M. Milakov and N. Gimelshein, "Online normalizer calculation for softmax," 2018, *arXiv:1805.02867*.
- [26] L. Liu, Z. Qu, Z. Chen, F. Tu, Y. Ding, and Y. Xie, "Dynamic sparse attention for scalable transformer acceleration," *IEEE Trans. Comput.*, vol. 71, no. 12, pp. 3165–3178, Dec. 2022.
- [27] T. J. Ham et al., "ELSA: Hardware-software co-design for efficient, lightweight self-attention mechanism in neural networks," in *Proc. ACM/IEEE 48th Annu. Int. Symp. Comput. Archit. (ISCA)*, 2021, pp. 692–705.
- [28] L. Lai, N. Suda, and V. Chandra, "CMSIS-NN: Efficient neural network kernels for ARM cortex-M CPUs," 2018, *arXiv:1801.06601*.
- [29] S. G. Bhaskaracharya, J. Demouth, and V. Grover, "Automatic kernel generation for Volta tensor cores," 2020, *arXiv:2006.12645*.
- [30] C. L. Lawson, R. J. Hanson, D. R. Kincaid, and F. T. Krogh, "Basic linear algebra subprograms for Fortran usage," *ACM Trans. Math. Softw.*, vol. 5, no. 3, pp. 308–323, Sep. 1979.
- [31] M. Abadi et al., "TensorFlow: A system for large-scale machine learning," in *Proc. 12th USENIX Conf. Operating Syst. Des. Implementation (OSDI)*, USA: USENIX Association, 2016, pp. 265–283.
- [32] A. Ivanov, N. Dryden, T. Ben-Nun, S. Li, and T. Hoefler, "Data movement is all you need: A case study on optimizing transformers," in *Proc. Mach. Learn. Syst.*, vol. 3, 2021, pp. 711–732.
- [33] W. Liu, D. Anguelov, D. Erhan, C. Szegedy, S. Reed, C.-Y. Fu, and A. C. Berg, "SSD: Single shot MultiBox detector," in *Lecture Notes in Computer Science*. Springer International Publishing, 2016, pp. 21–37.
- [34] L. Mei, P. Houshmand, V. Jain, S. Giraldo, and M. Verhelst, "ZigZag: Enlarging joint architecture-mapping design space exploration for DNN accelerators," *IEEE Trans. Comput.*, vol. 70, no. 8, pp. 1160–1174, Aug. 2021.
- [35] L. Mei, K. Goetschalckx, A. Symons, and M. Verhelst, "DeFiNES: Enabling fast exploration of the depth-first scheduling space for DNN accelerators through analytical modeling," in *Proc. IEEE Int. Symp. High-Perform. Comput. Archit. (HPCA)*, Piscataway, NJ, USA: IEEE Press, 2023, pp. 570–583.
- [36] M. Alwani, H. Chen, M. Ferdman, and P. Milder, "Fused-layer CNN accelerators," in *Proc. 49th Annu. IEEE/ACM Int. Symp. Microarchit. (MICRO)*, 2016, pp. 1–12.
- [37] P. Busia et al., "Reducing false alarms in wearable seizure detection with EEGformer: A compact transformer model for MCUs," *IEEE Trans. Biomed. Circuits Syst.*, vol. 18, no. 3, pp. 608–621, Jun. 2024.
- [38] E. Frantar, S. Ashkboos, T. Hoefler, and D. Alistarh, "OPTQ: Accurate quantization for generative pre-trained transformers," in *Proc. 11th Int. Conf. Learn. Representations*, 2023, pp. 153–169.
- [39] A. Dosovitskiy et al., "An image is worth 16x16 words: Transformers for image recognition at scale," in *Proc. 9th Int. Conf. Learning Representations (ICLR)*, Virtual Event, Austria, OpenReview.net, 2021, pp. 307–329.

Original p. 16 is not reproduced in this excerpt.
Full article: IEEE Xplore, <https://doi.org/10.1109/TC.2024.3500360> (© IEEE).